MSG No. 223.108 $Pm'\bar{3}'n'$ [Type III, cubic]

Table 1: Wyckoff site: 2a, site symmetry: m'-3'.

No.	position	mapping
1	$[0,\ 0,\ 0]$	[1,8,9,10,17,18,19,20,21,22,23,24,25,32,33,34,41,42,43,44,45,46,47,48]
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[2,3,4,5,6,7,11,12,13,14,15,16,26,27,28,29,30,31,35,36,37,38,39,40]

Table 2: Wyckoff site: 6b, site symmetry: m'm'm'..

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	[1,8,9,10,25,32,33,34]
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	[2,3,13,14,26,27,37,38]
3	$\left[0,\ 0,\ \frac{1}{2}\right]$	[4,5,15,16,28,29,39,40]
4	$\left[0,\ \frac{1}{2},\ 0\right]$	[6,7,11,12,30,31,35,36]
5	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[17,20,22,24,41,44,46,48]
6	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[18,19,21,23,42,43,45,47]

Table 3: Wyckoff site: 6c, site symmetry: -4'm'.2

No.	position	mapping
1	$\left[\frac{1}{4},\ 0,\ \frac{1}{2}\right]$	[1,8,13,14,26,27,33,34]
2	$\left[\frac{3}{4},\ 0,\ \frac{1}{2}\right]$	[2,3,9,10,25,32,37,38]
3	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	[4,16,18,21,29,39,43,47]
4	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	[5,15,19,23,28,40,42,45]
5	$\left[\frac{1}{2},\ \frac{3}{4},\ 0\right]$	[6,11,20,22,31,36,41,48]
6	$\left[\frac{1}{2},\ \frac{1}{4},\ 0\right]$	[7,12,17,24,30,35,44,46]

Table 4: Wyckoff site: 6d, site symmetry: -4'm'.2

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{2},\ 0\right]$	[1,8,13,14,26,27,33,34]
2	$\left[\frac{3}{4},\ \frac{1}{2},\ 0\right]$	[2,3,9,10,25,32,37,38]
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	[4,16,18,21,29,39,43,47]
4	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	[5,15,19,23,28,40,42,45]
5	$\left[0,\ \frac{3}{4},\ \frac{1}{2}\right]$	[6,11,20,22,31,36,41,48]
6	$\left[0,\ \frac{1}{4},\ \frac{1}{2}\right]$	[7,12,17,24,30,35,44,46]

Table 5: Wyckoff site: 8e, site symmetry: .32

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	[1,12,14,16,17,18]
2	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	[2,7,9,15,19,24]
3	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	[3,4,10,11,20,21]
4	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	[5,6,8,13,22,23]

continued ...

Table 5

No.	position	mapping
5	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[25,36,38,40,41,42]
6	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[26,31,33,39,43,48]
7	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[27,28,34,35,44,45]
8	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[29,30,32,37,46,47]

Table 6: Wyckoff site: 12f, site symmetry: 2m'm'..

No.	position	mapping
1	$[x, 0, 0]$	[1,8,33,34]
2	$\left[x + \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[2,3,37,38]
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - x\right]$	[4,16,29,39]
4	$\left[\frac{1}{2}, \frac{1}{2}, x + \frac{1}{2}\right]$	[5,15,28,40]
5	$\left[\frac{1}{2}, x + \frac{1}{2}, \frac{1}{2}\right]$	[6,11,31,36]
6	$\left[\frac{1}{2}, \frac{1}{2} - x, \frac{1}{2}\right]$	[7,12,30,35]
7	$[-x, 0, 0]$	[9,10,25,32]
8	$\left[\frac{1}{2} - x, \frac{1}{2}, \frac{1}{2}\right]$	[13,14,26,27]
9	$[0, x, 0]$	[17,24,44,46]
10	$[0, 0, x]$	[18,21,43,47]
11	$[0, 0, -x]$	[19,23,42,45]
12	$[0, -x, 0]$	[20,22,41,48]

Table 7: Wyckoff site: 12g, site symmetry: 2m'm'..

No.	position	mapping
1	$\left[x, 0, \frac{1}{2}\right]$	[1,8,33,34]
2	$\left[x + \frac{1}{2}, 0, \frac{1}{2}\right]$	[2,3,37,38]
3	$\left[0, \frac{1}{2}, \frac{1}{2} - x\right]$	[4,16,29,39]
4	$\left[0, \frac{1}{2}, x + \frac{1}{2}\right]$	[5,15,28,40]
5	$\left[\frac{1}{2}, x + \frac{1}{2}, 0\right]$	[6,11,31,36]
6	$\left[\frac{1}{2}, \frac{1}{2} - x, 0\right]$	[7,12,30,35]
7	$\left[-x, 0, \frac{1}{2}\right]$	[9,10,25,32]
8	$\left[\frac{1}{2} - x, 0, \frac{1}{2}\right]$	[13,14,26,27]
9	$\left[\frac{1}{2}, x, 0\right]$	[17,24,44,46]
10	$\left[0, \frac{1}{2}, x\right]$	[18,21,43,47]
11	$\left[0, \frac{1}{2}, -x\right]$	[19,23,42,45]
12	$\left[\frac{1}{2}, -x, 0\right]$	[20,22,41,48]

Table 8: Wyckoff site: 12h, site symmetry: 2m'm'..

No.	position	mapping
1	$\left[x, \frac{1}{2}, 0\right]$	[1,8,33,34]
2	$\left[x + \frac{1}{2}, \frac{1}{2}, 0\right]$	[2,3,37,38]
3	$\left[\frac{1}{2}, 0, \frac{1}{2} - x\right]$	[4,16,29,39]

continued ...

Table 8

No.	position	mapping
4	$\left[\frac{1}{2}, 0, x+\frac{1}{2}\right]$	[5,15,28,40]
5	$\left[0, x+\frac{1}{2}, \frac{1}{2}\right]$	[6,11,31,36]
6	$\left[0, \frac{1}{2}-x, \frac{1}{2}\right]$	[7,12,30,35]
7	$\left[-x, \frac{1}{2}, 0\right]$	[9,10,25,32]
8	$\left[\frac{1}{2}-x, \frac{1}{2}, 0\right]$	[13,14,26,27]
9	$\left[0, x, \frac{1}{2}\right]$	[17,24,44,46]
10	$\left[\frac{1}{2}, 0, x\right]$	[18,21,43,47]
11	$\left[\frac{1}{2}, 0, -x\right]$	[19,23,42,45]
12	$\left[0, -x, \frac{1}{2}\right]$	[20,22,41,48]

Table 9: Wyckoff site: 16i, site symmetry: .3.

No.	position	mapping
1	$[x, x, x]$	[1,17,18]
2	$\left[x+\frac{1}{2}, \frac{1}{2}-x, x+\frac{1}{2}\right]$	[2,7,15]
3	$\left[x+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}-x\right]$	[3,4,11]
4	$\left[\frac{1}{2}-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[5,6,13]
5	$[x, -x, -x]$	[8,22,23]
6	$[-x, x, -x]$	[9,19,24]
7	$[-x, -x, x]$	[10,20,21]
8	$\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[12,14,16]
9	$[-x, -x, -x]$	[25,41,42]
10	$\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{2}-x\right]$	[26,31,39]
11	$\left[\frac{1}{2}-x, \frac{1}{2}-x, x+\frac{1}{2}\right]$	[27,28,35]
12	$\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[29,30,37]
13	$[-x, x, x]$	[32,46,47]
14	$[x, -x, x]$	[33,43,48]
15	$[x, x, -x]$	[34,44,45]
16	$\left[x+\frac{1}{2}, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[36,38,40]

Table 10: Wyckoff site: 24j, site symmetry: ..2

No.	position	mapping
1	$\left[\frac{1}{4}, y, y+\frac{1}{2}\right]$	[1,13]
2	$\left[\frac{3}{4}, -y, y+\frac{1}{2}\right]$	[2,10]
3	$\left[\frac{3}{4}, y, \frac{1}{2}-y\right]$	[3,9]
4	$\left[y, y+\frac{1}{2}, \frac{1}{4}\right]$	[4,18]
5	$\left[-y, y+\frac{1}{2}, \frac{3}{4}\right]$	[5,19]
6	$\left[\frac{1}{2}-y, \frac{3}{4}, y\right]$	[6,20]
7	$\left[y+\frac{1}{2}, \frac{1}{4}, y\right]$	[7,17]
8	$\left[\frac{1}{4}, -y, \frac{1}{2}-y\right]$	[8,14]
9	$\left[y+\frac{1}{2}, \frac{3}{4}, -y\right]$	[11,22]
10	$\left[\frac{1}{2}-y, \frac{1}{4}, -y\right]$	[12,24]
11	$\left[y, \frac{1}{2}-y, \frac{3}{4}\right]$	[15,23]

continued ...

Table 10

No.	position	mapping
12	$\left[-y,\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	[16,21]
13	$\left[\frac{3}{4},\ -y,\ \frac{1}{2}-y\right]$	[25,37]
14	$\left[\frac{1}{4},\ y,\ \frac{1}{2}-y\right]$	[26,34]
15	$\left[\frac{1}{4},\ -y,\ y+\frac{1}{2}\right]$	[27,33]
16	$\left[-y,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	[28,42]
17	$\left[y,\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	[29,43]
18	$\left[y+\frac{1}{2},\ \frac{1}{4},\ -y\right]$	[30,44]
19	$\left[\frac{1}{2}-y,\ \frac{3}{4},\ -y\right]$	[31,41]
20	$\left[\frac{3}{4},\ y,\ y+\frac{1}{2}\right]$	[32,38]
21	$\left[\frac{1}{2}-y,\ \frac{1}{4},\ y\right]$	[35,46]
22	$\left[y+\frac{1}{2},\ \frac{3}{4},\ y\right]$	[36,48]
23	$\left[-y,\ y+\frac{1}{2},\ \frac{1}{4}\right]$	[39,47]
24	$\left[y,\ y+\frac{1}{2},\ \frac{3}{4}\right]$	[40,45]

Table 11: Wyckoff site: 24k, site symmetry: m'..

No.	position	mapping
1	$[0,\ y,\ z]$	[1,32]
2	$\left[\frac{1}{2},\ \frac{1}{2}-z,\ y+\frac{1}{2}\right]$	[2,27]
3	$\left[\frac{1}{2},\ z+\frac{1}{2},\ \frac{1}{2}-y\right]$	[3,26]
4	$\left[z+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[4,40]
5	$\left[\frac{1}{2}-z,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[5,39]
6	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[6,35]
7	$\left[y+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[7,36]
8	$[0,\ -y,\ -z]$	[8,25]
9	$[0,\ y,\ -z]$	[9,34]
10	$[0,\ -y,\ z]$	[10,33]
11	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[11,30]
12	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[12,31]
13	$\left[\frac{1}{2},\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$	[13,38]
14	$\left[\frac{1}{2},\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$	[14,37]
15	$\left[z+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[15,29]
16	$\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[16,28]
17	$[z,\ 0,\ y]$	[17,48]
18	$[y,\ z,\ 0]$	[18,45]
19	$[-y,\ z,\ 0]$	[19,47]
20	$[-z,\ 0,\ y]$	[20,46]
21	$[-y,\ -z,\ 0]$	[21,42]
22	$[z,\ 0,\ -y]$	[22,44]
23	$[y,\ -z,\ 0]$	[23,43]
24	$[-z,\ 0,\ -y]$	[24,41]

Table 12: Wyckoff site: `48l`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[x+\frac{1}{2},\ \frac{1}{2}-z,\ y+\frac{1}{2}\right]$	`[2]`
3	$\left[x+\frac{1}{2},\ z+\frac{1}{2},\ \frac{1}{2}-y\right]$	`[3]`
4	$\left[z+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[4]`
5	$\left[\frac{1}{2}-z,\ y+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[5]`
6	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[6]`
7	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[7]`
8	$[x,\ -y,\ -z]$	`[8]`
9	$[-x,\ y,\ -z]$	`[9]`
10	$[-x,\ -y,\ z]$	`[10]`
11	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[11]`
12	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	`[12]`
13	$\left[\frac{1}{2}-x,\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$	`[13]`
14	$\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$	`[14]`
15	$\left[z+\frac{1}{2},\ \frac{1}{2}-y,\ x+\frac{1}{2}\right]$	`[15]`
16	$\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ \frac{1}{2}-x\right]$	`[16]`
17	$[z,\ x,\ y]$	`[17]`
18	$[y,\ z,\ x]$	`[18]`
19	$[-y,\ z,\ -x]$	`[19]`
20	$[-z,\ -x,\ y]$	`[20]`
21	$[-y,\ -z,\ x]$	`[21]`
22	$[z,\ -x,\ -y]$	`[22]`
23	$[y,\ -z,\ -x]$	`[23]`
24	$[-z,\ x,\ -y]$	`[24]`
25	$[-x,\ -y,\ -z]$	`[25]`
26	$\left[\frac{1}{2}-x,\ z+\frac{1}{2},\ \frac{1}{2}-y\right]$	`[26]`
27	$\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ y+\frac{1}{2}\right]$	`[27]`
28	$\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ x+\frac{1}{2}\right]$	`[28]`
29	$\left[z+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-x\right]$	`[29]`
30	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	`[30]`
31	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[31]`
32	$[-x,\ y,\ z]$	`[32]`
33	$[x,\ -y,\ z]$	`[33]`
34	$[x,\ y,\ -z]$	`[34]`
35	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[35]`
36	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[36]`
37	$\left[x+\frac{1}{2},\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$	`[37]`
38	$\left[x+\frac{1}{2},\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$	`[38]`
39	$\left[\frac{1}{2}-z,\ y+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[39]`
40	$\left[z+\frac{1}{2},\ y+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[40]`
41	$[-z,\ -x,\ -y]$	`[41]`
42	$[-y,\ -z,\ -x]$	`[42]`
43	$[y,\ -z,\ x]$	`[43]`
44	$[z,\ x,\ -y]$	`[44]`
45	$[y,\ z,\ -x]$	`[45]`
46	$[-z,\ x,\ y]$	`[46]`

continued ...

Table 12

No.	position	mapping
47	$[-y,\ z,\ x]$	`[47]`
48	$[z,\ -x,\ y]$	`[48]`